Technical Guide: Atari SIO2PC USB Interface

Project: Future-Proofing Atari 8-Bit Connectivity with AspeQt-2k26

Building an Atari SIO2PC cable is a straightforward and cost-effective way to connect your vintage Atari hardware to a modern PC running **Windows, Linux, or macOS**. By using a USB-to-TTL chipset, you can utilize the **AspeQt-2k26** emulator to serve disk images and manage files directly on real Atari hardware.

Our BBS and project updates can be found at: https://13leader.net

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1. Parts & Requirements

- **USB-to-TTL Adapter:** Common chipsets include the **PL2303TA** or the **FT232RL**.
- **Atari SIO Connector:** Required for the cable interface. These can be sourced from eBay or specialty vendors like Best Electronics.
- **AspeQt-2k26 Software:** Compatible with RespeQt and legacy AspeQt emulators.

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2. Build Option A: PL2303TA Chipset

This version is widely compatible with **Windows, Linux, and macOS** for standard disk emulation.

Wiring Specifications

Atari SIO Pin	Signal Description	PL2303TA Wire Color
Pin 3	Data In	**TXD (Green)**
Pin 4	Ground	**GND (Black)**
Pin 5	Data Out	**RXD (White)**

Software Setting: In AspeQt, set **Handshake** to **"None"**.

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3. Build Option B: FTDI FT232RL Chipset

The FTDI version is highly recommended for advanced users as it is also compatible with **APE Atarimax** software.

Wiring Specifications

Atari SIO Pin	Signal Description	FT232RL Pin
Pin 3	Data In	**TXD**
Pin 4	Ground	**GND**
Pin 5	Data Out	**RXD**
Pin 7	Command Signal	**CTS**

Software Setting: In AspeQt, set **Handshake** to **"CTS"**.

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4. Software Connectivity (All Platforms)

1. **Driver Installation:** Ensure your OS recognizes the USB chipset (FTDI or Prolific).
2. **Launch AspeQt-2k26:** * **Serial Port:** Select the COM port (Windows) or /dev/ttyUSBx (Linux/macOS) associated with your cable.
    - **Baud Rate:** Standard start is **19,200**. Once verified, AspeQt-2k26 supports high-speed SIO routines.
3. **Integrated TNFS:** Use the built-in browser to pull files directly from the web to your Atari.

5. Troubleshooting

- **Handshake Mismatch:** If the cable fails to "see" the Atari, double-check your Handshake setting (None for PL2303, CTS for FTDI).
- **Wiring Check:** Ensure TXD and RXD are correctly swapped (Atari Data In connects to USB TXD).
- **OS Compatibility:** The FT232RL chipset generally has better driver support across modern macOS and Linux versions.
